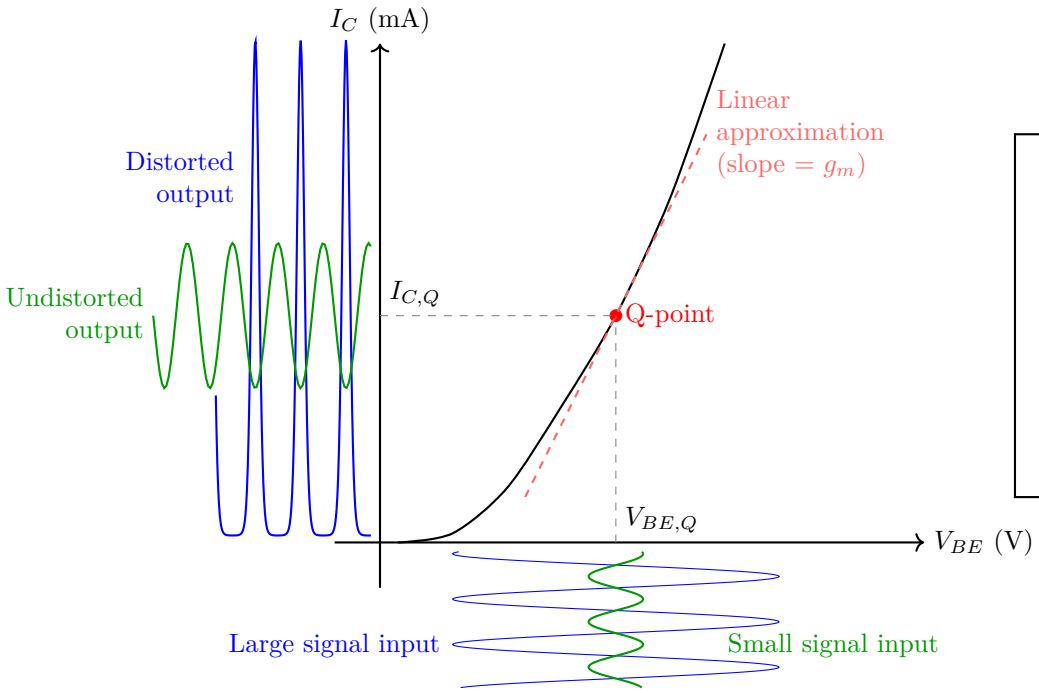
ELEC2104/9704 Tutorial 7 Solutions (2025)

Motivation: BJT Amplifiers

In this course, we're particularly interested in using BJTs as **amplifiers**. BJTs excel as amplifiers due to their exponential $I_C = I_S e^{V_{BE}/V_T}$ relationship, providing high transconductance ($g_m = I_C/V_T$). That is, small changes in base-emitter voltage V_{BE} lead to large changes in collector current I_C.

Small Signal Inputs vs. Large Signal Inputs

However, the very property that makes BJTs good amplifiers (the exponential relationship) also creates a challenge: distortion!
The I_C-V_{BE} curve is highly nonlinear, as shown in Figure 1. Large input signals cause the transistor to operate over a wide range of this curve, leading to distortion in the output signal (see in blue).



Key Insight:

Large signals see the full exponential ⇒ Distortion!

Small signals see only linear region ⇒ Good amplifier!

Figure 1: BJT nonlinearity: Large signals experience distortion due to the exponential I_C-V_{BE} relationship, while small signals remain in the approximately linear region around the Q-point.

Small signal inputs (see in green) keep the transistor operating near a specific point on the curve, called the **Quiescent point** (**Q-point**). Around this point, we can approximate the exponential curve as a straight line (the tangent with slope = g_m), allowing for *linear amplification without distortion.*

Q-point location - We deliberately position the Q-point on the I_C-V_{BE} characteristic curve by selecting an appropriate DC collector current $I_{C,Q}$. This choice involves trade-offs:

- Higher $I_{C,Q}$ → Higher g_m → Better gain, but more power consumption

- Lower $I_{C,Q} \rightarrow$ Lower power, but reduced gain and bandwidth

As Figure 1 demonstrates, the Q-point acts as the "center" of our linear approximation. The tangent line (slope = g_m) at the Q-point defines how small signals are amplified without distortion. To set this Q-point, we use a special circuit topology called a **Biasing network**. We will see examples of these circuits in Q1 and Q2.

Small Signal Analysis

The distinction between small signals and large signals for BJTs leads us to a powerful technique for analyzing BJT amplifier circuits: **Small-signal analysis**!

This technique is based on **linearization around the Q-point**. Mathematically, we approximate the nonlinear exponential curve $I_C = I_S e^{V_{BE}/V_T}$ with its first-order Taylor expansion around the Q-point:

$$I_C \approx I_{C,Q} + g_m \cdot (V_{BE} - V_{BE,Q})$$

where $g_m = \frac{dI_C}{dV_{BE}}\big|_Q = \frac{I_{C,Q}}{V_T}$ is the transconductance (slope of the tangent line in Figure 1).

This linear approximation is *only valid for small signals* because higher-order terms in the Taylor series become significant for large deviations from the Q-point. The beauty: this transforms complex nonlinear circuit analysis into straightforward linear circuit analysis using familiar techniques like Ohm's law and voltage dividers!

This linearization enables us to create the **hybrid-π small-signal model**—a linear equivalent circuit that replaces the nonlinear BJT for small-signal analysis:

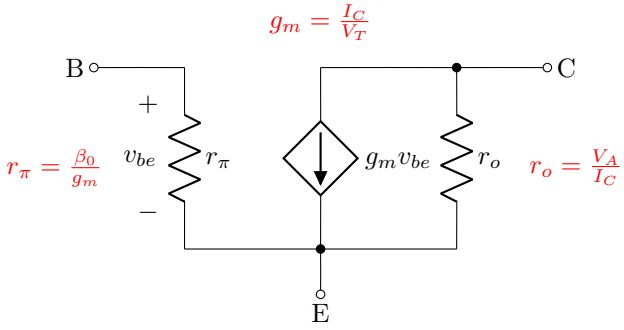


Figure 2: Hybrid-π small-signal model: The linear equivalent circuit that replaces the nonlinear BJT for small-signal analysis. All parameters depend on the Q-point!

Small Signal Analysis vs. Large Signal Analysis

In this course, we approach BJT amplifier analysis by following two sequential analysis techniques:

1. **Large-Signal (DC) Analysis**: First, we find the DC operating point (yet another name for the Q-point) in the absence of any AC signals. This tells us where the transistor is "sitting" when no signal is applied. Here, "DC" means constant (time-invariant) voltages and currents.
2. **Small-Signal (AC) Analysis**: Second, we study how the circuit responds to small AC signals *superimposed* onto the DC operating point. Here, "AC" refers to **any time-varying signal** (not just sinusoidal!) that is small enough to remain in the linear region around the Q-point, e.g. square waves, triangular waves, audio signals, etc.

Notation Convention

Throughout this course, we use a consistent notation to distinguish DC and AC quantities:

- **Capital letters** for DC (large-signal) quantities: V_C, I_B, V_{BE}, I_C, etc.
- **Lowercase letters** for AC (small-signal) quantities: v_c, i_b, v_{be}, i_c, etc.

Q-Point: Bridging Between Large and Small

The crucial connection between these two analysis steps is that **we need to know I_C (the DC collector current) to build our small-signal model**. The transconductance $g_m = I_C/V_T$ is the fundamental parameter that determines how the BJT amplifies signals. Without knowing the DC operating point, we cannot determine the small-signal behavior! This is why we must perform large-signal analysis first.

Note that the Q-point $Q = (I_C, V_{CE})$ is expressed with the collector current I_C and the collector-emitter voltage V_{CE}. Only I_C directly influences the small-signal parameters like g_m, r_π, and r_o. The voltage V_{CE} is primarily used to verify that the BJT is operating in the forward active region (i.e. $V_{CE} > V_{BE}$ for npn BJTs).

Also, **note that the V_{BE} voltage is not part of the Q-point**, but knowing the Q-point guarantees knowing V_{BE}.

BJT Analysis Procedure

The standard procedure for analyzing BJT amplifier circuits focuses on finding the two key quantities: collector current I_C and collector-emitter voltage V_{CE}. Assume forward active region, then:

Step 1: Large-Signal (DC) Analysis

1. Find I_C and V_{CE} using standard circuit analysis techniques (may need Thevenin equivalence).
   - Use KCL/KVL to write equations for the circuit (typically 2x KVL loop equations).
   - Use the exponential relationship to relate V_{BE} to I_C: $V_{BE} = V_T \ln(I_C/I_S)$ (no Early Effect).
   - If you end up with equation terms of I_B, use $I_C = \beta_0 I_B$ to express these in terms of I_C.
   - For terms of I_E, use $I_E = I_C + I_B = I_C(1 + 1/\beta_0)$ to express in terms of I_C.
2. **Verify the forward active assumption:** Check that $V_{BE,\text{on}} \leq V_{BE} \leq V_{CE}$
3. Record Q-point: $Q = (I_C, V_{CE})$

Step 2: Small-Signal (AC) Analysis

1. Calculate small-signal parameters from Q-point:

$$\text{transconductance } g_m = \frac{I_C}{V_T} \tag{1}$$

$$\pi\text{-resistance } r_\pi = \frac{\beta_0}{g_m} \tag{2}$$

$$\text{output resistance } r_o = \frac{V_A}{I_C} \tag{3}$$

2. Replace the BJT circuit symbol with the linearised hybrid-π model within the circuit diagram.
3. Set all DC independent sources to zero (current sources become open circuits, voltage sources become short circuits).
4. Analyze the resulting circuit diagram using standard linear circuit analysis techniques.
   - If the amplifier is a common-emitter topology (as in most questions), you should find that (with collector resistor R_C and load resistor R_L):

$$\text{input impedance } R_{in} = r_\pi \tag{4}$$

$$\text{output impedance } R_{out} = r_o \parallel R_C \tag{5}$$

$$\text{voltage gain } A_v = -g_m \cdot (R_C \parallel r_o \parallel R_L) \tag{6}$$

   - **More on amplifier topologies and parameters in the following tutorials!**

In this course, you'll rarely encounter problems where the BJT operates in saturation or cutoff (but it is still possible) — we focus on the amplifying region!

Critical Assumption: Forward Active Region Only!

IMPORTANT: Both small-signal and large-signal analysis techniques we use in this course are **only valid in the forward active region** of the BJT!

The exponential relationship $I_C = I_S e^{V_{BE}/V_T}$ is not a universal law—it's simply a mathematical approximation that models a BJT operating in forward active mode. Outside this region (cutoff, saturation), the relationship breaks down completely.

Check as a *rule of thumb* that your calculated voltages satisfy $V_C > V_B > V_E$ (npn) or $V_E > V_B > V_C$ (pnp).

BJT Non-Ideal Behavior: Early Effect

In our previous analysis, we assumed the ideal BJT relationship $I_C = I_S e^{V_{BE}/V_T}$, where the collector current depends only on the base-emitter voltage V_{BE}. However, real BJTs exhibit a non-ideal phenomenon called the **Early Effect**, where the collector current also depends on the collector-emitter voltage V_{CE}.

Early Effect in Large-Signal (DC) Analysis

Early Effect is described in the mathematical model by including a V_{CE} term:

$$I_C = I_S e^{V_{BE}/V_T} \left(1 + \frac{V_{CE}}{V_A}\right)$$

where V_A is the Early voltage. This equation shows that:

- Higher $V_{CE} \Rightarrow$ higher I_C (for fixed V_{BE})
- When $V_A \to \infty$ (infinite Early voltage), we recover the ideal model.

The Early Effect also affects the current gain:

$$\beta_F = \beta_{F0} \left(1 + \frac{V_{CE}}{V_A}\right)$$

where β_{F0} is the zero-bias current gain.

Early Effect in Small-Signal (AC) Models

In small-signal analysis, the Early Effect manifests as a **finite output resistance** r_o in the hybrid-π model. Recall from Figure 2 that we include this resistance between collector and emitter. This resistance is given (approximated) by:

$$r_o = \frac{V_A}{I_C}$$

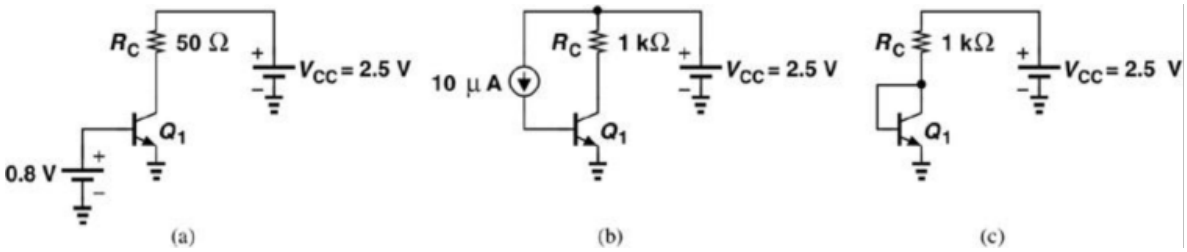
Infinite Early voltage ($V_A \to \infty$) gives infinite output resistance (ideal current source).

Questions will assume $V_A = \infty$, an infinite Early voltage (i.e. no Early Effect), unless otherwise specified.

1 Question 1 Solution

1.1 Part A

Determine the operating point and the small-signal model of Q_1 for each of the circuits shown below. Assume $I_S = 8 \times 10^{-16}$ A, $V_T = 26$ mV, $\beta = 100$, $V_A = \infty$.



Recall the typical steps for large-signal analysis of a BJT circuit:

1. Find I_C and V_{CE} using standard circuit analysis techniques (may need Thevenin equivalence).
   - Use KCL/KVL to write equations for the circuit (typically 2x KVL loop equations).
   - Use the non-linear I_C-V_{BE} relationship to relate V_{BE} to I_C (no Early Effect!).
   - If you end up with equation terms of I_B, use $I_C = \beta_0 I_B$ to express these in terms of I_C.
   - For terms of I_E, use $I_E = I_C + I_B = I_C(1 + 1/\beta_0)$ to express in terms of I_C.
2. **Verify the forward active assumption:** Check that $V_{BE,\text{on}} \leq V_{BE} \leq V_{CE}$
3. Record Q-point: $Q = (I_C, V_{CE})$

Note that $V_A = \infty$ in this question, so Early Effect does not apply!

Large-signal analysis:

In this circuit diagram, we notice that the BJT base is *directly* connected to a voltage source of 0.8 V. This is generous - we can directly derive V_{BE} which (without Early Effect) means we have I_C ✓✓
V_E is connected to ground, so $V_E = 0$ V :

$$V_{BE} = V_B - V_E = 0.8\,\text{V} - 0\,\text{V} = 0.8\,\text{V} \tag{1}$$

Using the relationship between I_C and V_{BE} (no Early Effect) and the known V_{BE}, we can determine:

$$I_C = I_S \cdot \exp\left(\frac{V_{BE}}{V_T}\right) = \left(8 \times 10^{-16}\,\text{A}\right) \exp\left(\frac{0.8\,\text{V}}{0.026\,\text{V}}\right) = 18.5\,\text{mA}$$

Finding V_{CE}: Using a KVL loop through R_C and the collector (or just Ohm's law):

$$V_C = V_{CC} - I_C R_C = 2.5\,\text{V} - (0.0185\,\text{A}) \cdot (50\,\Omega) = 1.58\,\text{V} \tag{2}$$

Then, we can determine V_{CE}.

$$V_{CE} = V_C - V_E = 1.58\,\text{V} - 0\,\text{V} = 1.58\,\text{V} \tag{3}$$

Sanity check for forward active region: $V_{BE,on} \leq V_{BE} \leq V_{CE}$. Assuming $V_{BE,on}$ is around 0.6 - 0.7 V, we have $0.7\,\text{V} \leq 0.8\,\text{V} \leq 1.58\,\text{V}$ ✓✓

Therefore, the Q-point of the BJT is:

$$Q = (I_C, V_{CE}) = (18.5\,\text{mA},\ 1.58\,\text{V})$$

Remark: Note that in this case, we did not need to use Thevenin equivalence or write 2x KVL equations because the circuit is simple in that V_{BE} was effectively given.

Small-signal analysis:

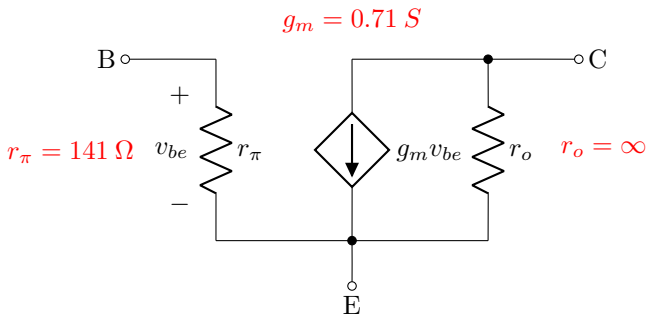
According to the given current gain $\beta = 100$ and the thermal voltage $V_T = 0.026\,\text{V}$, (and $V_A = \infty \Rightarrow$ no Early Effect) we can determine the following small signal parameters:

$$g_m = \frac{I_C}{V_T} = \frac{0.0185\,\text{A}}{0.026\,\text{V}} = 0.71\,\text{S} \tag{4}$$

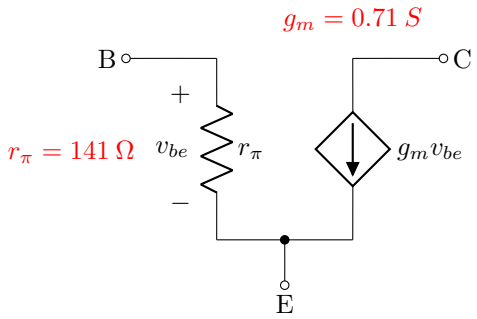
$$r_\pi = \frac{\beta}{g_m} = \frac{100}{0.71\,\text{S}} = 141\,\Omega \tag{5}$$

$$r_o = \frac{V_A}{I_C} = \frac{\infty}{I_C} = \infty \tag{6}$$

Finally, we draw the small signal model:



Note that an infinite resistance is equivalent to an open circuit. Thus, here, r_o can be omitted as it is infinite:



Important: KVL Loops with Ground Nodes

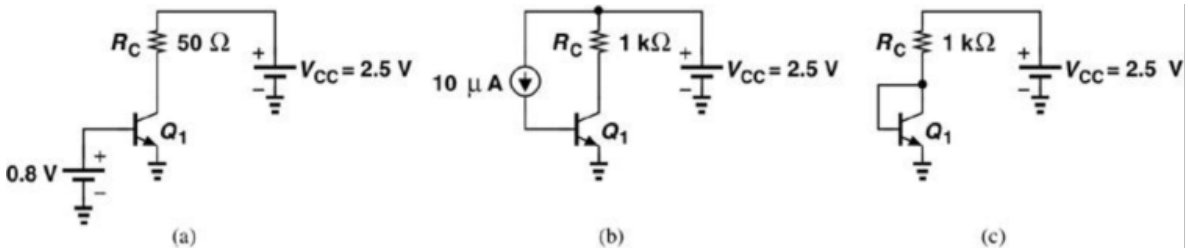
In circuit analysis with ground nodes, KVL loops don't look like visible "circles" anymore! The ground node acts as a common reference point that completes the loop implicitly.
Example from Part A:

- The KVL loop goes: $V_{CC} \to R_C \to$ Collector $\to$ Emitter $\to$ GND $\to V_{CC}$
- But we write it as: $V_{CC} - I_C R_C - V_{CE} = 0$, or simply $V_C = V_{CC} - I_C R_C$
- The ground connection "closes" the loop even though it's not drawn as a wire

Key insight: Every voltage source connected to ground has an implicit return path through ground. This is why we can write $V_C = V_{CC} - I_C R_C$ without explicitly showing the complete loop — the ground node provides the reference that makes this equation valid.

1.2 Part B

Determine the operating point and the small-signal model of Q_1 for each of the circuits shown below. Assume $I_S = 8 \times 10^{-16}\,\text{A}$, $V_T = 26\,\text{mV}$, $\beta = 100$, $V_A = \infty$.



Recall the typical steps for large-signal analysis of a BJT circuit:

1. Find I_C and V_{CE} using standard circuit analysis techniques (may need Thevenin equivalence).
   - Use KCL/KVL to write equations for the circuit (typically 2x KVL loop equations).
   - Use the non-linear I_C-V_{BE} relationship to relate V_{BE} to I_C (no Early Effect!).
   - If you end up with equation terms of I_B, use $I_C = \beta_0 I_B$ to express these in terms of I_C.
   - For terms of I_E, use $I_E = I_C + I_B = I_C(1 + 1/\beta_0)$ to express in terms of I_C.
2. **Verify the forward active assumption:** Check that $V_{BE,\text{on}} \leq V_{BE} \leq V_{CE}$
3. Record Q-point: $Q = (I_C, V_{CE})$

Note that $V_A = \infty$ in this question, so Early Effect does not apply!

Large-Signal Analysis

From the circuit diagram, a current source provides the base current I_B directly. As in Part A, we can directly determine I_C without KVL, this time using the current gain.

$$I_B = 10\,\mu\text{A} \tag{7}$$

Using the known current gain $\beta = 100$, the collector current I_C is determined:

$$I_C = \beta I_B = 100 \cdot (10 \times 10^{-6}\,\text{A}) = 1\,\text{mA} \tag{8}$$

Finding V_{CE}: We can use KVL through the collector to determine the collector-emitter voltage V_{CE}:

$$V_{CE} = V_{CC} - I_C R_C = 2.5\,\text{V} - (10^{-3}\,\text{A}) \cdot (1000\,\Omega) = 1.5\,\text{V} \tag{9}$$

Therefore, the Q-point of the BJT is:

$$Q = (I_C, V_{CE}) = (1\,\text{mA},\ 1.5\,\text{V}) \tag{10}$$

Now, we need to find V_{BE} to verify the forward active assumption. Using the I_C-V_{BE} relationship and the known I_C, we can determine V_{BE}:

$$V_{BE} = V_T \ln\left(\frac{I_C}{I_S}\right) = (0.026\,\text{V})\ \ln\left(\frac{10^{-3}\,\text{A}}{8 \times 10^{-16}\,\text{A}}\right) = 0.724\,\text{V} \qquad [V_{BE,on} \leq V_{BE} \leq V_{CE}\ \checkmark\checkmark] \tag{11}$$

Small-Signal Analysis

The transconductance is calculated as:

$$g_m = \frac{I_C}{V_T} = \frac{10^{-3}\,\text{A}}{0.026\,\text{V}} = 0.0385\,\text{S} \tag{12}$$

The π-resistance is:

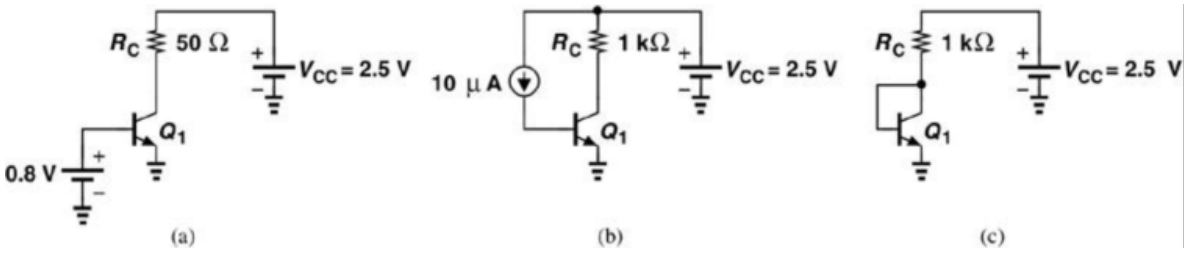
$$r_\pi = \frac{\beta}{g_m} = \frac{100}{0.0385\,\text{S}} = 2600\,\Omega \tag{13}$$

The Early voltage resistance is:

$$r_o = \frac{V_A}{I_C} = \frac{\infty}{I_C} = \infty \tag{14}$$

1.3 Part C

Determine the operating point and the small-signal model of Q_1 for each of the circuits shown below. Assume $I_S = 8 \times 10^{-16}$ A, $V_T = 26$ mV, $\beta = 100$, $V_A = \infty$.



Recall the typical steps for large-signal analysis of a BJT circuit:

1. Find I_C and V_{CE} using standard circuit analysis techniques (may need Thevenin equivalence).
   - Use KCL/KVL to write equations for the circuit (typically 2x KVL loop equations).
   - Use the non-linear I_C-V_{BE} relationship to relate V_{BE} to I_C (no Early Effect!).
   - If you end up with equation terms of I_B, use $I_C = \beta_0 I_B$ to express these in terms of I_C.
   - For terms of I_E, use $I_E = I_C + I_B = I_C(1 + 1/\beta_0)$ to express in terms of I_C.
2. **Verify the forward active assumption:** Check that $V_{BE,\text{on}} \leq V_{BE} \leq V_{CE}$
3. Record Q-point: $Q = (I_C, V_{CE})$

Note that $V_A = \infty$ in this question, so Early Effect does not apply!

Large-Signal Analysis

Unfortunately, for this circuit, we cannot directly determine I_C or V_{BE} from the circuit as we have not been given a convenient voltage/current source this time, so we need to write 2x KVL or KCL equations and use the BJT I_C-V_{BE} relationship to solve for I_C.

The first thing that stands out is the base and collector of the BJT are tied together. This means 3 things:

- The voltages at these nodes are equal: $V_B = V_C \;\Rightarrow\; V_{BE} = V_{CE}$
- Therefore, the BJT is "self-biased" - the BJT is in the active region as long as $V_{BE} > V_{BE,on}$.
- By KCL, the current flowing through resistor R_C is simply the emitter current $I_E = I_C + I_B$.

You should know that the emitter current equals the sum of the collector current and the base current:

$$I_E = I_C + I_B = I_C \left(1 + \frac{1}{\beta}\right) \tag{15}$$

Here, we have expressed I_E in terms of I_C using the current gain, $I_B = I_C/\beta$

Using KVL through the collector resistor R_C (or more simply, Ohm's law), we can write down another expression for I_E :

$$V_{CC} - I_{R_C} R_C = V_{CE} \quad \Rightarrow \quad I_{R_C} = \frac{V_{CC} - V_{CE}}{R_C} \tag{16}$$

Note that $I_{R_C} = I_E$, as by KCL at the collector node, $I_{R_C} = I_C + I_B = I_E$, so we now have:

$$I_E = \frac{V_{CC} - V_{CE}}{R_C} \tag{17}$$

Combining these two expressions for I_E, we get the following:

$$I_C\left(1+\frac{1}{\beta}\right)=\frac{V_{CC}-V_{CE}}{R_C} \tag{18}$$

We still need to express I_C in terms of known quantities. Recall, as mentioned earlier, that since the base and collector are tied together, $V_{BE}=V_{CE}$. We can use the BJT I_C-V_{BE} relationship to express $V_{CE}=V_{BE}$ in terms of I_C:

$$\begin{aligned} I_C\left(1+\frac{1}{\beta}\right)&=\frac{V_{CC}-V_{BE}}{R_C}\\ V_{BE}&=V_T\ln\left(\frac{I_C}{I_S}\right)\\ I_C&=\frac{\beta}{1+\beta}\cdot\frac{V_{CC}-V_T\ln\left(\frac{I_C}{I_S}\right)}{R_C} \end{aligned} \tag{19}$$

The above implicit equation cannot be solved analytically for I_C. We can instead use numerical methods, such as the Newton Iteration Method.

Newton Iteration Method

Define the function for iteration:

$$\frac{\beta}{1+\beta}\cdot\frac{V_{CC}-V_T\ln\left(\frac{I_C}{I_S}\right)}{R_C}-I_C=0 \tag{20}$$

$$f(I_C)=\frac{\beta}{1+\beta}\cdot\frac{V_{CC}-V_T\ln\left(\frac{I_C}{I_S}\right)}{R_C}-I_C \tag{21}$$

The derivative is

$$f'(I_C)=-\frac{\beta}{1+\beta}\cdot\frac{V_T}{R_C\cdot I_C}-1 \tag{22}$$

Then the Newton update rule is

$$I_C^{(n+1)}=I_C^{(n)}-\frac{f(I_C^{(n)})}{f'(I_C^{(n)})} \tag{23}$$

Substituting the given values ($\beta=100$, $V_{CC}=2.5\,\mathrm{V}$, $R_C=1000\,\Omega$, $V_T=0.026\,\mathrm{V}$, $I_S=8\times10^{-16}\,\mathrm{A}$):

$$f(I_C)=\frac{100}{101}\cdot\frac{2.5-0.026\ln\left(\frac{I_C}{8\times10^{-16}}\right)}{1000}-I_C \tag{24}$$

Starting with an initial guess $I_C^{(0)}=1\,\mathrm{mA}$:

Iteration	$I_C^{(n)}$ (mA)	$f(I_C^{(n)})$	$f'(I_C^{(n)})$
0	1.000	7.58×10^{-4}	-1.026
1	1.739	4.78×10^{-6}	-1.015
2	1.744	9.43×10^{-11}	-1.015
3	1.744		

Using this method, the collector current converges to

$$I_C\approx1.74\,\mathrm{mA} \tag{25}$$

The collector–emitter voltage equals the base–emitter voltage, hence

$$V_{CE}=V_{BE}=V_T\ln\left(\frac{I_C}{I_S}\right)=(0.026\,\mathrm{V})\ln\left(\frac{1.74\times10^{-3}\,\mathrm{A}}{8\times10^{-16}\,\mathrm{A}}\right)=0.739\,\mathrm{V}\qquad[V_{BE,on}\le V_{BE}\le V_{CE},\ \text{self-biased}\ \checkmark\checkmark] \tag{26}$$

Thus, the Q-point is

$$Q=(I_C,V_{CE})=(1.74\,\mathrm{mA},\ 0.739\,\mathrm{V}) \tag{27}$$

Small-Signal Parameters

The transconductance is

$$g_m = \frac{I_C}{V_T} = \frac{1.74 \times 10^{-3}\,\text{A}}{0.026\,\text{V}} = 66.92\,\text{mS} \tag{28}$$

The π-resistance is

$$r_\pi = \frac{\beta}{g_m} = \frac{100}{0.06692\,\text{S}} \approx 1494\,\Omega \tag{29}$$

Since $V_A = \infty$, the early-effect resistance is

$$r_o = \infty \tag{30}$$

Simpler Iteration Method (Gauss-Seidel Method)

For BJT circuits where the transistor is on, there's a simpler iteration method that often converges without derivatives!

The Method: Rearrange the equation to isolate I_C on the left side:

$$I_C = \frac{\beta}{1+\beta} \cdot \frac{V_{CC} - V_T \ln(I_C/I_S)}{R_C}$$

Then iteratively compute:

$$I_C^{(n+1)} = \frac{\beta}{1+\beta} \cdot \frac{V_{CC} - V_T \ln(I_C^{(n)}/I_S)}{R_C}$$

Calculator Technique:

1. Enter initial guess: `1E-3` (for 1 mA)
2. Type the formula using `ANS` for $I_C^{(n)}$:

   `(100/101)*(2.5-0.026*ln(ANS/8E-16))/1000`

3. Press `=` repeatedly until the value converges

Example for Part C:

Press =	Display (mA)
Initial	1.000
1st	1.758
2nd	1.744
3rd	1.744

Warning: This method may not converge if the initial guess is too far from the solution or if the BJT is near cutoff.

2 Question 2

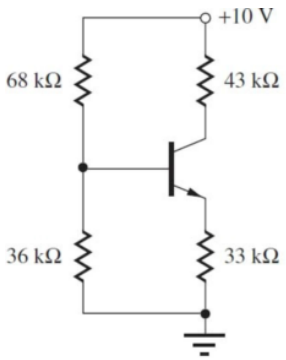
(a) Find the Q-point for the circuit in Fig. a. Assume that $\beta_F = 50$ and $V_{BE} = 0.7$ V.

(b) Repeat the calculation if all the resistor values are decreased by a factor of 5.

(c) Repeat if all the resistor values are increased by a factor of 5.

(d) Find the Q-point in part (a) using the numerical iteration method if $I_S = 0.5$ fA and $V_T = 25.8$ mV.

(e) Repeat for $V_A = 60$ V.



(a)

Solution:

This circuit is a "8"-shaped **4-Resistor Bias Network** BJT circuit. This is a well known configuration, featured in the lecture. You should memorise this, or draw it on your formula sheet!
We must simplify the left-hand-side part of the circuit using Thevenin equivalence (see lecture).
The equivalent voltage source can be determined by finding the open-circuit voltage at the base node:

$$V_{eq} = 10 \cdot \left(\frac{36\,\text{k}\Omega}{36\,\text{k}\Omega + 68\,\text{k}\Omega} \right) = 3.46\,\text{V} \tag{1}$$

The equivalent resistance can be determined by:

$$R_{eq} = 36\,\text{k}\Omega \parallel 68\,\text{k}\Omega = 23.54\,\text{k}\Omega \tag{2}$$

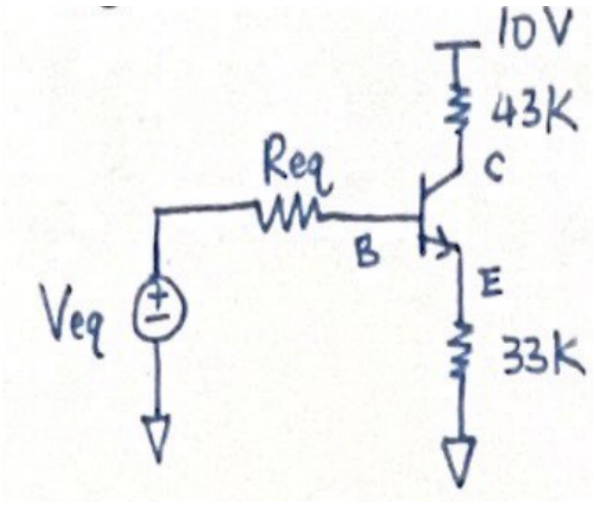


Figure 3: Q2.Thevenin equivalent circuit

2.1 Part A

Applying KVL in the base loop, we get the following equation.

$$V_{\text{eq}} - I_B R_{\text{eq}} - V_{BE} - I_E \cdot 33\,000 = 0 \tag{3}$$

The correlation between the emitter current and the base current is given by:

$$I_E = (1 + \beta_F) I_B \tag{4}$$

Substituting (2) into (1), we get the following equation, where $V_{BE} = 0.7V$, given by the question.

$$V_{\text{eq}} - I_B R_{\text{eq}} - V_{BE} - (1 + \beta_F) I_B \times 33\,000 = 0 \tag{5}$$

Rearranging the equation, we can solve the base current.

$$I_B = \frac{V_{\text{eq}} - V_{BE}}{R_{\text{eq}} + (1 + \beta_F) \times 33\,000} I_B = 1.62\ \mu\text{A} \tag{6}$$

The collector current and the emitter current are then determined using the current gain $\beta_F = 50$.

$$I_C = \beta_F I_B = 50 \times 1.62\ \mu\text{A} = 80.87\ \mu\text{A} \tag{7}$$

$$I_E = (1 + \beta_F) I_B = 51 \times 1.62\ \mu\text{A} = 82.48\ \mu\text{A} \tag{8}$$

According to Ohm's Law and basic KVL, we can get the expressions for the collector voltage and emitter voltage, where The collector–emitter voltage:

$$V_C = V_{CC} - I_C \cdot 43\,000 \tag{9}$$

$$V_E = I_E \cdot 33\,000 \tag{10}$$

Thus we can get the voltage drop between the collector and the emitter.

$$V_{CE} = V_C - V_E = \big[V_{CC} - (43\,000 \cdot I_C)\big] - \big(33\,000 \cdot I_E\big) = 3.8\,\text{V} \tag{11}$$

Therefore, the operating point (Q-point) is:

$$Q : \big(I_C, V_{CE}\big) = \big(80.87\ \mu\text{A},\ 3.8\ \text{V}\big) \tag{12}$$

2.2 Part B

From the voltage divider, the Thevenin equivalent voltage is given as:

$$V_{\text{eq}} = 3.46\ \text{V} \tag{13}$$

The Thevenin resistance is the parallel combination of 7.2 kΩ and 13.6 kΩ:

$$R_{\text{eq}} = 7.2\ \text{k}\Omega \parallel 13.6\ \text{k}\Omega = 4.71\ \text{k}\Omega \tag{14}$$

The base current can be expressed as:

$$I_B = \frac{V_{\text{eq}} - V_{BE}}{R_{\text{eq}} + (1 + \beta_F) \cdot 3300} \tag{15}$$

Substituting the given values ($V_{BE} = 0.7$ V, $\beta_F = 50$), we obtain:

$$I_B = \frac{3.46 - 0.7}{4710 + 51 \times 330} = 8.10\ \mu\text{A} \tag{16}$$

The collector current and emitter current are then calculated:

$$I_C = \beta_F I_B = 50 \times 8.10\ \mu\text{A} = 404.6\ \mu\text{A} \tag{17}$$

$$I_E = (1 + \beta_F) I_B = 51 \times 8.10\ \mu\text{A} = 412.7\ \mu\text{A} \tag{18}$$

Next, we determine the collector–emitter voltage:

$$V_{CE} = \big[10 - (8.6\text{k}\Omega \cdot I_C)\big] - (6.6\text{k}\Omega \cdot I_E) = 3.8\ \text{V} \tag{19}$$

Therefore, the operating point (Q-point) is:

$$Q : \big(I_C, V_{CE}\big) = \big(405\ \mu\text{A},\ 3.8\ \text{V}\big) \tag{20}$$

2.3 Part C

From the voltage divider, the Thevenin equivalent voltage is the same as before:

$$V_{\text{eq}} = 3.46 \text{ V} \tag{21}$$

The Thevenin resistance is the parallel combination of 180 kΩ and 340 kΩ:

$$R_{\text{eq}} = 180 \text{ k}\Omega \parallel 340 \text{ k}\Omega = 117.7 \text{ k}\Omega \tag{22}$$

The base current is then:

$$I_B = \frac{V_{\text{eq}} - V_{BE}}{R_{\text{eq}} + (1+\beta_F) \cdot 3300 \cdot 5} \tag{23}$$

Substituting the given values ($V_{BE} = 0.7$ V, $\beta_F = 50$), we obtain:

$$I_B = \frac{3.46 - 0.7}{117700 + 51 \times 16500} = 0.32 \ \mu\text{A} \tag{24}$$

The collector current and emitter current are then determined:

$$I_C = \beta_F I_B = 50 \times 0.32 \ \mu\text{A} = 16.2 \ \mu\text{A} \tag{25}$$

$$I_E = (1+\beta_F) I_B = 51 \times 0.32 \ \mu\text{A} = 16.5 \ \mu\text{A} \tag{26}$$

The collector–emitter voltage is:

$$V_{CE} = 3.8 \text{ V} \tag{27}$$

Therefore, the operating point (Q-point) is:

$$Q : \big(I_C, V_{CE}\big) = \big(16.2 \ \mu\text{A},\ 3.8 \text{ V}\big) \tag{28}$$

2.4 Part D

The base–emitter voltage is modeled using the following equation:

$$V_{BE} = V_T \cdot \ln\left(\frac{I_C}{I_S} + 1\right) = 0.0258 \cdot \ln\big(2 \times 10^{15} I_C + 1\big) \tag{29}$$

The collector current can also be expressed using the KVL method:

$$I_C = \frac{V_{\text{eq}} - V_{BE}}{R_{\text{eq}} + (1+\beta_F) R_E} \cdot \beta_F \tag{30}$$

Substituting the known values, we can get the expression of V_{BE} as a function of I_C:

$$I_C = \frac{3.46 - V_{BE}}{170.654\text{k}\Omega} \cdot 50 = \frac{3.46 - V_{BE}}{34130} \tag{31}$$

$$V_{BE} = 3.46 - 34130 I_C \tag{32}$$

To solve the non-linear equations (29) and (32), we construct the following function. We want to find the optimal value of I_C to make $f(I_C) = 0$

$$f(I_C) = V_{BE@(eq.29)} - V_{BE@(eq.32)} = 34130 I_C + 0.0258 \ln\big(2 \times 10^{15} I_C + 1\big) - 3.46 \tag{33}$$

The derivative of the function is:

$$f'(I_C) = 34130 + \frac{0.0258 \cdot 2 \times 10^{15}}{2 \times 10^{15} I_C + 1} \tag{34}$$

Using the Newton's iteration method:

$$I_{C,\text{new}} = I_{C,\text{old}} - \frac{f(I_{C,\text{old}})}{f'(I_{C,\text{old}})} \tag{35}$$

After iteration, the solution converges to:

$$I_C = 81.8 \ \mu\text{A}, \qquad V_{CE} = 3.79 \text{ V} \tag{36}$$

Therefore, the operating point (Q-point) is:

$$Q : \big(I_C, V_{CE}\big) = \big(81.8 \ \mu\text{A},\ 3.79 \text{ V}\big) \tag{37}$$

2.5 Part E

Including the Early effect, the forward current gain is expressed as:

$$\beta_F = \beta_{F0}\left(1 + \frac{V_{CE}}{V_A}\right) = 50 \cdot \left(1 + \frac{V_{CE}}{60}\right) \tag{38}$$

The collector–emitter voltage drop can be written below using KVL.

$$V_{CE} = (V_{CC} - I_C R_C) - I_E R_E \tag{39}$$

Since $I_C = \beta_F I_B$ and $I_E = (1 + \beta_F) I_B$, we can rewrite the equation (38) to be:

$$V_{CE} = V_{CC} - \Big[I_B \cdot \beta_F\Big] R_C - \Big[I_B \cdot (\beta_F + 1)\Big] R_E \tag{40}$$

Meanwhile, the base loop KVL gives:

$$V_{\text{eq}} - R_{\text{eq}} I_B - V_{BE} - \Big[I_B \cdot (\beta_F + 1)\Big] R_E = 0 \tag{41}$$

Equations (39) and (40) are solved simultaneously. The numerical results are:

$$V_{CE} = 3.44 \text{ V}, \qquad I_B = 0.81 \ \mu\text{A}, \qquad I_C = 85.5 \ \mu\text{A} \tag{42}$$

Therefore, the operating point (Q-point) is:

$$Q : \big(I_C, V_{CE}\big) = \big(85.5 \ \mu\text{A},\ 3.44 \text{ V}\big) \tag{43}$$

3 Question 3

An npn transistor is operating in the forward-active region with a base current of $3\,\mu$A. It is found that $I_C = 240\,\mu$A for $V_{CE} = 5\,$V and $I_C = 265\,\mu$A for $V_{CE} = 10\,$V. What are the values of β_{F0} and V_A for this transistor?

Solution:

We define the following parameters:

$$I_B = \text{ Base current } = 3\,\mu\text{A} = 3 \times 10^{-6}\,\text{A} \tag{1}$$

$$\beta_F = \text{ Forward common-emitter current gain} \tag{2}$$

$$\beta_{F0} = \text{ Forward common-emitter current gain at } V_{CE} = 0 \tag{3}$$

$$V_A = \text{ Early voltage} \tag{4}$$

$$V_{CE} = \text{ Collector–emitter voltage} \tag{5}$$

$$I_C = \text{ Collector current} \tag{6}$$

We have the equation that describes how V_{CE} affects I_C

$$I_C = \beta_F I_B = \left[\beta_{F0}\left(1 + \frac{V_{CE}}{V_A}\right)\right] I_B \tag{7}$$

Taking the two sets of values of V_{CE} and I_C into (5), we can get the equation set.

$$240\,\mu\text{A} = \beta_{F0}\left(1 + \frac{5}{V_A}\right)\ (3\,\mu\text{A}) \tag{8}$$

$$265\,\mu\text{A} = \beta_{F0}\left(1 + \frac{10}{V_A}\right)\ (3\,\mu\text{A}) \tag{9}$$

Solving (6) and (7), we get

$$\beta_{F0} = 71.7, \qquad V_A = 43\,\text{V}. \tag{10}$$